

## 11-JEE TEST # 3

M.M.: 300

TIME: 3 HOURS

### Test Syllabus:

**Physics: Vectors and Projectile 3D**

**Chemistry: Chemical Bonding and Periodic Table**

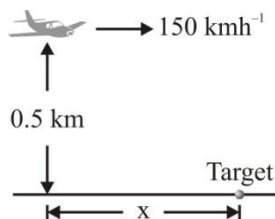
**Mathematics: Binomial Theorem and Logarithm**

### General Instructions:

1. Duration of Test is 3 hrs.
2. The Test booklet consists of 75 questions. The maximum marks are 300.
3. There are three parts in the question paper consisting of Physics, Chemistry and Mathematics.
4. Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
5. Use blue/black ballpoint pen only to darken the appropriate circle.
6. Mark should be dark and completely fill the circle.
7. Dark only one circle for each entry.
8. Dark the circle in the space provided only.
9. Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

### PHYSICS

1. The aeroplane is shown here on level flight at an altitude of 0.5 km and at a speed of  $150 \text{ kmh}^{-1}$ .



At what distance  $x$  should it release a bomb to hit the target? [Take  $g = 10 \text{ ms}^{-2}$ ]

- (a) 150 m                      (b) 420 m                      (c) 2000 m                      (d) 800 m
2. Mark out the incorrect statements for plane to plane projectile motion.
- (a) Horizontal component of the velocity always remains constant.  
 (b) Minimum speed of the projectile is at its highest point and is non-zero.  
 (c) Speed of the projectile remains constant.  
 (d) Speed of the projectile at the time of launching and landing are same.
3. The two projectiles would have same maximum height if their
- (a) Time of flight are the same  
 (b) Ranges are the same  
 (c) Vertical components of initial velocity are the same  
 (d) Horizontal components of initial velocity are the same.
4. If the horizontal range is  $4\sqrt{3}$  times of its maximum height, the angle of projection is
- (a)  $45^\circ$                       (b)  $30^\circ$                       (c)  $60^\circ$                       (d)  $75^\circ$
5. A projectile is projected at an initial velocity,  $\mathbf{v} = (a\hat{i} + b\hat{j})$  and range is double of its height, then
- (a)  $b = \frac{a}{2}$                       (b)  $b = a$                       (c)  $b = 2a$                       (d)  $b = 3a$ .



6. A ball is dropped from a moving train along a horizontal straight track. The path of the ball observed by an observer standing on ground is  
 (a) Circular (b) Hyperbolic (c) Straight line (d) Parabolic
7. A body is projected from the top of tower with horizontal velocity  $4 \text{ ms}^{-1}$ . The velocity of the body after  $0.7 \text{ s}$  is  
 (a)  $15 \text{ ms}^{-1}$  (b)  $10 \text{ ms}^{-1}$  (c)  $9.5 \text{ ms}^{-1}$  (d)  $8 \text{ ms}^{-1}$
8. The bodies are projected at angle  $30^\circ$  and  $60^\circ$  w.r.t. horizontal. The ratio of their maximum height attained is (for same range)  
 (a)  $1 : 3$  (b)  $1 : 1$  (c)  $2 : 1$  (d)  $1 : 4$
9. The equation of trajectory is given by  $y = \sqrt{3}x - \frac{gx^2}{2}$ . The angle of projection is  
 (a) Zero (b)  $\frac{\pi}{2}$  (c)  $\theta = \tan^{-1} \sqrt{3}$  (d)  $\theta = \tan^{-1} \frac{1}{\sqrt{3}}$
10. If the range of a particle projected with certain velocity at  $15^\circ$  angle is  $1.5 \text{ km}$ , what will be its range when angle is  $45^\circ$  with same initial velocity  
 (a)  $0.75 \text{ km}$  (b)  $1.5 \text{ km}$  (c)  $3 \text{ km}$  (d)  $6 \text{ km}$
11. Three vectors  $\vec{P}$ ,  $\vec{Q}$ , and  $\vec{R}$  are such that  $|\vec{P}| = |\vec{Q}|$ ,  $|\vec{R}| = \sqrt{2}|\vec{P}|$ , and  $\vec{P} + \vec{Q} + \vec{R} = 0$ . The angles between  $\vec{P}$  and  $\vec{Q}$ ,  $\vec{Q}$  and  $\vec{R}$  and  $\vec{P}$  and  $\vec{R}$  are  
 (a)  $90^\circ, 135^\circ, 135^\circ$  (b)  $90^\circ, 45^\circ, 45^\circ$  (c)  $45^\circ, 90^\circ, 90^\circ$  (d)  $45^\circ, 135^\circ, 135^\circ$
12. If  $\vec{A} = 4\hat{i} - 2\hat{j} + 6\hat{k}$  and  $\vec{B} = \hat{i} - 2\hat{j} - 3\hat{k}$ , the angle made by  $\vec{A} + \vec{B}$  with the x-axis is  
 (a)  $\cos^{-1}\left(\frac{3}{\sqrt{50}}\right)$  (b)  $\cos^{-1}\left(\frac{4}{\sqrt{50}}\right)$  (c)  $\cos^{-1}\left(\frac{5}{\sqrt{50}}\right)$  (d)  $\cos^{-1}\sqrt{\frac{12}{50}}$
13. Two vectors with magnitude 3 units and 4 units are given. The resultant these vectors 1 unit. The value of the dot product of the two vectors is  
 (a) 0 (b)  $-1$  unit (c)  $+5$  units (d)  $-12$  units
14. The sum of magnitudes of two forces acting at a point is  $16 \text{ N}$ . If their resultant is normal to the smaller force and has a magnitude of  $8 \text{ N}$ . Then the forces are:  
 (a)  $6 \text{ N}, 10 \text{ N}$  (b)  $8 \text{ N}, 8 \text{ N}$  (c)  $4 \text{ N}, 12 \text{ N}$  (d)  $2 \text{ N}, 14 \text{ N}$ .
15. If two vectors are equal and their resultant is also equal to one of them, then the angle between the two vectors is:  
 (a)  $60^\circ$  (b)  $120^\circ$  (c)  $90^\circ$  (d)  $0^\circ$
16.  $|\vec{A} \times \vec{B}| = \sqrt{3}\vec{A} \cdot \vec{B}$ , then the value of  $|\vec{A} + \vec{B}|$  is:  
 (a)  $\left(A^2 + B^2 + \frac{AB}{\sqrt{3}}\right)^{1/2}$  (b)  $A + B$  (c)  $\left(A^2 + B^2 + \sqrt{3}AB\right)^{1/2}$  (d)  $\left(A^2 + B^2 + AB\right)^{1/2}$
17. Which of the following is the unit vector perpendicular to  $\vec{A}$  and  $\vec{B}$ ?  
 (a)  $\frac{\hat{A} \times \hat{B}}{AB \sin \theta}$  (b)  $\frac{\hat{A} \times \hat{B}}{AB \cos \theta}$  (c)  $\frac{\vec{A} \times \vec{B}}{AB \sin \theta}$  (d)  $\frac{\vec{A} \times \vec{B}}{AB \cos \theta}$
18. If the vectors  $\vec{P} = a\hat{i} + a\hat{j} + 3\hat{k}$  and  $\vec{Q} = a\hat{i} - 2\hat{j} - \hat{k}$  are perpendicular to each other then the positive value of  $a$  is:  
 (a) Zero (b) 1 (c) 2 (d) 3
19. The component of vector  $\vec{A} = 2\vec{i} + 3\vec{j}$  along the vector  $\vec{i} + \vec{j}$  is:  
 (a)  $\frac{5}{\sqrt{2}}$  (b)  $10\sqrt{2}$  (c)  $5\sqrt{2}$  (d) 5



20. The area of parallelogram represented by the vectors  $\vec{A} = 2\vec{i} + 3\vec{j}$  and  $\vec{B} = \vec{i} + 4\vec{j}$  is:  
 (a) 14 unit (b) 7.5 unit (c) 10 unit (d) 5 unit
21. A particle A is projected from the ground with an initial velocity of  $10 \text{ ms}^{-1}$  at an angle  $60^\circ$  with horizontal. From what height should another particle B be projected horizontally with velocity  $5 \text{ ms}^{-1}$  so that both the particles collide in ground at point C if both are projected simultaneously  $g = 10 \text{ ms}^{-2}$ .  
 (a) 10 m (b) 15 m (c) 20 m (d) 30 m
22. Two stones are projected with the same velocity in magnitude but making different angles with the horizontal. Their ranges are equal. If the angle of projection of one is  $\pi/3$  its maximum height is  $y_1$ , the maximum height of the other will be  
 (a)  $3y_1$  (b)  $2y_1$  (c)  $\frac{y_1}{2}$  (d)  $\frac{y_1}{3}$
23. A body is projected at angle  $30^\circ$  with Kinetic Energy E. The K.E. at the top pt. of its journey is  
 (a)  $\frac{E}{2}$  (b)  $\frac{E}{3}$  (c)  $\frac{3E}{4}$  (d) Zero.
24. The trajectory of a projectile in a vertical plane is  $y = ax - bx^2$ , where a and b are constants and x and y respectively the horizontal and vertical distances of the projectile from the point of projection. The maximum height attained is:  
 (a)  $\frac{2a^2}{b}$  (b)  $\frac{a^2}{b}$  (c)  $\frac{a^2}{2b}$  (d)  $\frac{a^2}{4b}$
25. A projectile is fired horizontally with a velocity u from the top of a hill. What should be the time, in which projectile should hit the ground so that the velocity with which projectile hits the ground makes an angle of  $45^\circ$  with the horizontal:  
 (a)  $\frac{2u}{g}$  (b)  $\frac{u}{g}$  (c)  $\frac{u}{2g}$  (d)  $\frac{u\sqrt{2}}{2g}$

## CHEMISTRY

26. According to the Periodic Law of elements, the variation in properties of elements is related to their  
 (a) nuclear masses  
 (b) atomic numbers  
 (c) nuclear neutron-proton number ratios  
 (d) atomic masses
27. Which one of the following is an amphoteric oxide ?  
 (a)  $\text{Na}_2\text{O}$  (b)  $\text{SO}_2$  (c)  $\text{B}_2\text{O}_3$  (d)  $\text{ZnO}$
28. Which one of the following ions has the highest value of ionic radius ?  
 (a)  $\text{O}^{2-}$  (b)  $\text{B}^{3+}$  (c)  $\text{Li}^+$  (d)  $\text{F}^-$
29. In which of the following arrangements, the order is NOT according to the property indicated against it?  
 (a)  $\text{Li} < \text{Na} < \text{K} < \text{Rb}$  :  
     Increasing metallic radius  
 (b)  $\text{I} < \text{Br} < \text{F} < \text{Cl}$  :  
     Increasing electron gain enthalpy  
     (with negative sign)  
 (c)  $\text{B} < \text{C} < \text{N} < \text{O}$   
     Increasing first ionization enthalpy





Increasing ionic size

30. The correct sequence which shows decreasing order of the ionic radii of the elements is  
 (a)  $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+ > \text{F}^- > \text{O}^{2-}$  (b)  $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+} > \text{O}^{2-} > \text{F}^-$   
 (c)  $\text{Na}^+ > \text{F}^- > \text{Mg}^{2+} > \text{O}^{2-} > \text{Al}^{3+}$  (d)  $\text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$
31. Which of the following represents the correct order of increasing first ionization enthalpy for Ca, Ba, S, Se and Ar ?  
 (a)  $\text{Ca} < \text{S} < \text{Ba} < \text{Se} < \text{Ar}$  (b)  $\text{S} < \text{Se} < \text{Ca} < \text{Ba} < \text{Ar}$   
 (c)  $\text{Ba} < \text{Ca} < \text{Se} < \text{S} < \text{Ar}$  (d)  $\text{Ca} < \text{Ba} < \text{S} < \text{Se} < \text{Ar}$
32. The ionic radii (in Å) of  $\text{N}^{3-}$ ,  $\text{O}^{2-}$  and  $\text{F}^-$  are respectively :  
 (a) 1.71, 1.40 and 1.36 (b) 1.71, 1.36 and 1.40  
 (c) 1.36, 1.40 and 1.71 (d) 1.36, 1.71 and 1.40
33. Among the following, which compound will show the highest lattice energy ?  
 (a) KF (b) NaF (c) CsF (d) RbF.
34. The shape of  $\text{SF}_4$  molecule is  
 (a) Square planar (b) Trigonal bipyramidal  
 (c) Pyramidal (d) Distorted tetrahedron/see saw
35. Which of the following contains maximum number of lone pairs on the central atom ?  
 (a)  $\text{ClO}_3^-$  (b)  $\text{XeF}_4$  (c)  $\text{SF}_4$  (d)  $\text{I}_3^-$
36. The pair of species having identical shapes for molecules of both species is  
 (a)  $\text{XeF}_2, \text{CO}_2$  (b)  $\text{BF}_3, \text{PCl}_3$  (c)  $\text{PF}_5, \text{IF}_5$  (d)  $\text{CF}_4, \text{SF}_4$
37. Lattice energy of an ionic compound depends upon  
 (a) Charge on the ion and size of the ion (b) Packing of ions only  
 (c) Size of the ion only (d) Charge on the ion only
38. In which of the following molecules/ions are all the bonds not equal?  
 (a)  $\text{XeF}_4$  (b)  $\text{BF}_4^-$  (c)  $\text{SF}_4$  (d)  $\text{SiF}_4$
39. The hybridization of orbitals of N atom in  $\text{NO}_3^-$ ,  $\text{NO}_2^+$  and  $\text{NH}_4^+$  are respectively :  
 (a)  $sp, sp^2, sp^3$  (b)  $sp^2, sp, sp^3$  (c)  $sp, sp^3, sp^2$  (d)  $sp^2, sp^3, sp$
40. The structure of  $\text{IF}_7$  is  
 (a) square pyramidal (b) trigonal bipyramidal  
 (c) octahedral (d) pentagonal bipyramidal
41. In which of the following pairs the two species are not isostructural ?  
 (a)  $\text{CO}_3^{2-}$  and  $\text{NO}_3^-$  (b)  $\text{PCl}_4^+$  and  $\text{SiCl}_4$  (c)  $\text{PF}_5$  and  $\text{BrF}_5$  (d)  $\text{AlF}_6^{3-}$  and  $\text{SF}_6$
42. The species in which the N atom is in a state of  $sp$  hybridization is :  
 (a)  $\text{NO}_3^-$  (b)  $\text{NO}_2$  (c)  $\text{NO}_2^+$  (d)  $\text{NO}_2^-$
43. The first ionization energy (in kJ/mol) of Na, Mg, Al and Si respectively are  
 (a) 496, 577, 737, 786 (b) 786, 737, 577, 496  
 (c) 496, 577, 786, 737 (d) 496, 737, 577, 786



44. The five successive ionization enthalpies of an element are 800, 2427, 3658, 25024 and 32824 kJ mol<sup>-1</sup>. The number of valence electrons in the element is  
 (a) 3 (b) 4 (c) 2 (d) 5
45. The first ionization enthalpy of the following elements are in the order :  
 (a) C < N < Si < P (b) P < Si < N < C (c) P < Si < C < N (d) Si < P < C < N
46. Screening effect is not observed in  
 (a) He<sup>+</sup> (b) Li<sup>2+</sup> (c) Be<sup>3+</sup> (d) in all the three
47. In which case, effective nuclear charge is minimum ?  
 (a) Be (b) Be<sup>2+</sup>  
 (c) Be<sup>3+</sup> (d) all have the same effective nuclear charge.
48. Which of the following configuration represents atoms of the element having the highest second ionization potential ?  
 (a) 1s<sup>2</sup>2s<sup>2</sup>2p<sup>4</sup> (b) 1s<sup>2</sup>2s<sup>2</sup>2p<sup>6</sup> (c) 1s<sup>2</sup>2s<sup>2</sup>2p<sup>6</sup>3s<sup>1</sup> (d) 1s<sup>2</sup>2s<sup>2</sup>2p<sup>6</sup>3s<sup>2</sup>.
49. One mole of magnesium in the vapour state absorbed 1200 kJ mol<sup>-1</sup> of energy. If the first and second ionization energies of Mg are 750 and 1450 kJ mol<sup>-1</sup> respectively, the final composition of the mixture is  
 (a) 31% Mg<sup>+</sup> + 69% Mg<sup>2+</sup> (b) 69% Mg<sup>+</sup> + 31% Mg<sup>2+</sup>  
 (c) 86% Mg<sup>+</sup> + 14% Mg<sup>2+</sup> (d) 14% Mg<sup>+</sup> + 86% Mg<sup>2+</sup>
50. The second ionization energies of Li, Be, B and C are in the order  
 (a) Li > C > B > Be (b) Li > B > C > Be (c) B > C > Be > Li (d) Be > C > B > Li

### MATHEMATICS

51. The number of terms in the expansion of  $(1 + 5\sqrt{2}x)^9 + (1 - 5\sqrt{2}x)^9$  is –  
 (a) 5 (b) 7 (c) 9 (d) 10
52. The number of non zero terms in the expansion of  $[(1 + 3\sqrt{2}x)^9 - (1 - 3\sqrt{2}x)^9]$  is –  
 (a) 9 (b) 10 (c) 5 (d) 15
53. After simplification, the total number of terms in the expansion of  $(x + \sqrt{2})^4 + (x - \sqrt{2})^4$  is –  
 (a) 10 (b) 5 (c) 4 (d) 3
54. The number of terms in the expansion of  $[(x - 3y)^2(x + 3y)^2]^3$  is –  
 (a) 6 (b) 7 (c) 8 (d) None of these
55. The coefficient of  $x^4$  in the expansion of  $\left(\frac{x}{2} - \frac{3}{x^2}\right)^{10}$  is –  
 (a)  $\frac{405}{256}$  (b)  $\frac{504}{259}$  (c)  $\frac{450}{263}$  (d) None of these



56. The coefficient of  $x^{-26}$  in the expansion of  $\left(x^2 - \frac{2}{x^4}\right)^{11}$  is  
 (a)  $330 \times 2^6$  (b)  $-330 \times 2^6$  (c)  $330 \times 2^7$  (d)  $-330 \times 2^7$
57. The term independent of  $x$  in the expansion of  $\left(\sqrt{\frac{x}{3}} + \frac{3}{2x^2}\right)^{10}$  will be –  
 (a)  $3/2$  (b)  $5/4$  (c)  $5/2$  (d) None of these
58. The number of integral terms in the expansion of  $(5^{1/2} + 7^{1/6})^{642}$  is –  
 (a) 106 (b) 108 (c) 103 (d) 109
59. The middle term in the expansion of  $\left(x^3 - \frac{1}{x^3}\right)^{10}$  is –  
 (a) 252 (b) -252 (c) 210 (d) -210
60. If the middle term in the expansion of  $\left(x^2 + \frac{1}{x}\right)^n$  is  $924x^6$ , then  $n =$   
 (a) 10 (b) 12 (c) 14 (d) None of these
61. The greatest coefficient in the expansion of  $(1+x)^{10}$  is  
 (a)  $\frac{10!}{5!6!}$  (b)  $\frac{10!}{(5!)^2}$  (c)  $\frac{10!}{5!7!}$  (d) None of these
62. The 5th term from the end in the expansion of  $\left(\frac{x^3}{2} - \frac{2}{x^3}\right)^9$  is –  
 (a)  $63x^3$  (b)  $-\frac{252}{x^3}$  (c)  $\frac{672}{x^{18}}$  (d) None of these
63. If in the expansion of  $\left(2^{1/3} + \frac{1}{3^{1/3}}\right)^n$ , the ratio of 6th terms from beginning and from the end is  $1/6$ , then the value of  $n$  is –  
 (a) 5 (b) 7 (c) 9 (d) None of these
64. The value of  $\sqrt{(\log_{0.5}^2 4)}$  is  
 (a) -2 (b)  $\sqrt{-4}$  (c) 2 (d) None of these
65. The value of  $\log_3 4 \log_4 5 \log_5 6 \log_6 7 \log_7 8 \log_8 9$  is  
 (a) 1 (b) 2 (c) 3 (d) 4



66.  $\log_7 \log_7 \sqrt{7(\sqrt{7}\sqrt{7})} =$   
 (a)  $3\log_2 7$  (b)  $1-3\log_3 7$  (c)  $1-3\log_7 2$  (d) None of these
67. The value of  $81^{(1/\log_5 3)} + 27^{\log_9 36} + 3^{4/\log_7 9}$  is equal to  
 (a) 49 (b) 625 (c) 216 (d) 890
68. If  $\log_4 5 = a$  and  $\log_5 6 = b$ , then  $\log_3 2$  is equal to  
 (a)  $\frac{1}{2a+1}$  (b)  $\frac{1}{2b+1}$  (c)  $2ab+1$  (d)  $\frac{1}{2ab-1}$
69. If  $x = \log_a(bc)$ ,  $y = \log_b(ca)$ ,  $z = \log_c(ab)$ , then which of the following is equal to 1  
 (a)  $x+y+z$  (b)  $(1+x)^{-1} + (1+y)^{-1} + (1+z)^{-1}$   
 (c)  $x-y-z$  (d) None of these
70. The solution of the equation  $\log_7 \log_5 (\sqrt{x^2+5+x}) = 0$   
 (a)  $x=2$  (b)  $x=3$  (c)  $x=4$  (d)  $x=-2$
71. The value of  $(0.05)^{\log_{\sqrt{20}}(0.1+0.01+0.001+\dots)}$  is  
 (a) 81 (b)  $\frac{1}{81}$  (c) 20 (d)  $\frac{1}{20}$
72. If  $a, b, c$  are distinct positive numbers, each different from 1, such that  
 $[\log_b a \log_c a - \log_a a] + [\log_a b \log_c b - \log_b b] + [\log_a c \log_b c - \log_c c] = 0$ , then  $abc =$   
 (a) 1 (b) 2 (c) 3 (d) None of these
73. If  $\frac{\log x}{b-c} = \frac{\log y}{c-a} = \frac{\log z}{a-b}$ , then which of the following is false  
 (a)  $xyz=3$  (b)  $x^a y^b z^c = 1$  (c)  $x^{b+c} y^{c+a} z^{a+b} = 1$  (d)  $xyz = x^a y^b z^c$
74. If  $x^{\frac{3}{4}(\log_3 x)^2 + \log_3 x - \frac{5}{4}} = \sqrt{3}$  then  $x$  has not  
 (a) One positive integral value (b) One irrational value  
 (c) Two positive rational values (d) None of these
75. If  $\frac{1}{\log_3 \pi} + \frac{1}{\log_4 \pi} > x$ , then  $x$  be  
 (a) 2 (b) 3 (c) 3.5 (d)  $\pi$

**End of Paper**

